

Vinh Ly

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Summary

Software Engineer with **18 years** designing and shipping distributed systems at scale. Leads video processing infrastructure at YouTube handling billions of daily requests. Previously owned payment and search systems at Whatnot processing millions in daily transactions, and managed ML infrastructure teams at Google supporting petabyte-scale data pipelines. Go, **Python**, and C++ with on-call ownership of end-to-end systems. Partners with product and data science to translate complex requirements into incrementally valuable features. Owns production infrastructure end to end, including on call coverage, deployment automation, and monitoring across distributed services. Comfortable partnering with platform and application teams on reliability work.

Core Skills

Distributed Systems: Kubernetes, Terraform, gRPC, GraphQL, Redis, service mesh

Cloud Platforms: GCP, AWS, Docker, BigQuery, Spanner, Cloud Run

Languages & Frameworks: Go, Python, Java, C++, Apache Beam, Flask, TypeScript, JavaScript, HTML, CSS

Data Engineering: Kafka, Flink, Bigtable, PostgreSQL, MongoDB, Elasticsearch

Observability & ML Infra: Prometheus, Grafana, Datadog, TensorFlow, Kubeflow, MLflow

Work Experience

Staff Software Engineer | YouTube | 02/2025 - Present

- Designs video processing infrastructure handling billions of daily requests. Owns end-to-end pipeline from upload through transcoding and delivery, maintaining **SLA** compliance across **GCP** workloads with on-call rotation.
- Builds internal tooling for content moderation and classification at scale. Partners with ML teams to deploy **TensorFlow** models serving real-time inference with **sub-50ms P99 latency**.
- Implements cost optimization initiatives reducing cloud spend by 15%. Migrates legacy **Java** services to **Go** microservices on **Cloud Run**, improving deployment frequency and resource utilization.
- Drives cross-team initiatives to standardize API contracts and reduce service coupling. Introduces **GraphQL** federation layer unifying disparate data sources for frontend consumption.

Staff Software Engineer | Whatnot | 05/2024 - 02/2025

- Built payment transaction infrastructure processing over \$1M daily across marketplace transactions. Designed **PostgreSQL** schema and **Go** service layer handling checkout, refunds, and seller payouts with strong consistency guarantees.
- Implemented search infrastructure serving sub-100ms latency for millions of users. Designed caching strategy using **Redis** and **Elasticsearch** that reduced backend load by 40% and improved conversion metrics.
- Deployed comprehensive monitoring using **Prometheus** and **Grafana**. Created SLO-based alerting that reduced alert noise by 60%, improving on-call efficiency and system reliability.
- Conducted technical interviews and evaluated candidates for engineering roles. Structured interview loops focusing on system design and coding proficiency to maintain hiring bar.

Staff Software Engineer/Manager | Google | 03/2015 - 04/2024

- Managed team of 8 engineers maintaining ML training infrastructure processing petabytes of data daily. Owned end-to-end pipelines from **BigQuery** ingestion through **TensorFlow** model deployment with on-call accountability.
- Designed load balancing system for ML inference serving that improved throughput by 25%. Implemented **C++** proxy layer distributing requests across **GPU** clusters with health-aware routing.

- Built data pipeline handling 10TB daily for search ranking experiments. Used **Apache Beam** and **Dataflow** to process training data with sub-hour freshness for ranking model updates.
- Mentored engineers on distributed systems design and career development. Promoted 3 engineers to senior level and established technical growth frameworks for the team.

Software Engineer | Google | 02/2011 - 02/2015

- Optimized indexing pipeline reducing processing time by 30%. Profiled **C++** code using **perf** and **Valgrind** to identify bottlenecks in document parsing and inverted index construction.
- Developed backend services for search infrastructure handling millions of queries per second. Implemented **C++** components for result ranking and filtering with strict latency budgets.
- Participated in on-call rotation for search serving infrastructure. Resolved incidents affecting query latency and availability, developing runbooks for common failure patterns.
- Collaborated with search quality team to launch ranking algorithm changes. Implemented A/B testing framework for online experiments with statistical significance calculations.

Software Design Engineer II | Microsoft | 10/2007 - 12/2011

- Contributed to core platform systems used by thousands of developers internally. Wrote **C++** code for low-level system components with focus on performance and correctness.
- Debugged complex issues in production systems using kernel-level tracing and profiling. Resolved memory corruption and concurrency bugs in high-throughput server applications.
- Participated in design reviews and code reviews for critical system changes. Collaborated with senior engineers on architecture decisions affecting platform reliability.

Software Design Engineer Intern | Microsoft | 06/2006 - 08/2006

- Developed internal tooling for test automation using **C#** and **.NET** frameworks. Reduced manual testing effort for QA team through scripted test execution.
- Documented system components and contributed to technical specifications. Worked closely with full-time engineers to understand large-scale software development practices.

Education

B.S. Honors in Computer Science

Virginia Tech | 2004 - 2007